

Practical 7:

DOM Traversal and Manipulation

What is the DOM?

The **Document Object Model (DOM)** represents the structure of an HTML document as a tree of objects. It allows JavaScript to access, modify, or create HTML elements dynamically.

Common DOM Methods:

Method	Description
getElementById(id)	Selects an element by its ID.
querySelector(selector)	Selects the first element that matches a CSS selector.
createElement(tagName)	Creates a new HTML element.
appendChild(node)	Adds a new child node to a parent element.
removeChild(node)	Removes a child node from a parent element.

Syntax Explanation

// Accessing elements

```
const title = document.getElementById('title');  
const button = document.querySelector('.btn');
```

// Creating elements

```
const newItem = document.createElement('li');
```

// Adding content

```
newItem.textContent = "New Task";
```

// Adding to the DOM

```
document.body.appendChild(newItem);
```

// Removing an element

```
parentElement.removeChild(childElement);
```

Simple Examples

Example 1: Change Text Using getElementById

```
<h1 id="main-title">Hello</h1>  
<script>
```

```
document.getElementById('main-title').textContent = 'Welcome to DOM!';  
</script>
```

Example 2: Create and Append Element

```
<ul id="task-list"></ul>  
<script>  
  const li = document.createElement('li');  
  li.textContent = 'Buy groceries';  
  document.getElementById('task-list').appendChild(li);  
</script>
```

Example 3: Use querySelector to Change Style

```
<p class="note">This is a note.</p>  
<script>  
  const note = document.querySelector('.note');  
  note.style.color = 'blue';  
</script>
```

Example 4: Add Multiple Items with Button Click

```
<button onclick="addTask()">Add Task</button>  
<ul id="list"></ul>  
  
<script>  
  function addTask() {  
    const li = document.createElement('li');  
    li.textContent = 'New Task';  
    document.getElementById('list').appendChild(li);  
  }  
</script>
```

Example 5: Remove Element from DOM

```
<ul id="myList">  
  <li id="item1">Item 1</li>  
</ul>  
<script>  
  const item = document.getElementById('item1');
```

```
    item.parentNode.removeChild(item);  
</script>
```

Example 6: Toggle Task Done

```
<ul id="tasks">  
  <li onclick="this.classList.toggle('done')">Click me to mark done</li>  
</ul>
```

```
<style>  
  .done {  
    text-decoration: line-through;  
  }  
</style>
```

Tasks

Task 1: Add Items to a List

Create an input field and a button.

When the button is clicked, add the input value to a list.

Task 2: Delete Items from a List

Add a delete button beside each task.

Clicking it removes that task.

Task 3: Edit a Task

Add an "Edit" button beside each task.

Clicking it allows the user to change the task content.